

Protect Your Appliances With This Power-On Delay Timer

S. Thangamani

This simple delay timer circuit is used to prevent noise and spike in electrical equipment before switching on the main input supply. Most electrical appliances powered by 230V AC mains produce large voltage spikes when the power is switched on. The voltage spike also affects low-power electronic circuits. Hence, noise suppression and spike protection are very important. One of the easiest solutions is to use a power-on delay circuit that switches on the electronic circuit after a few seconds.

The author's prototype is shown in Fig. 1. Block diagram of the power-on delay timer circuit is shown in Fig. 2.

Circuit description

Circuit diagram of power-on delay timer circuit is shown in Fig. 3. It is built around a timer NE555 (IC1),



Fig. 1: Author's prototype

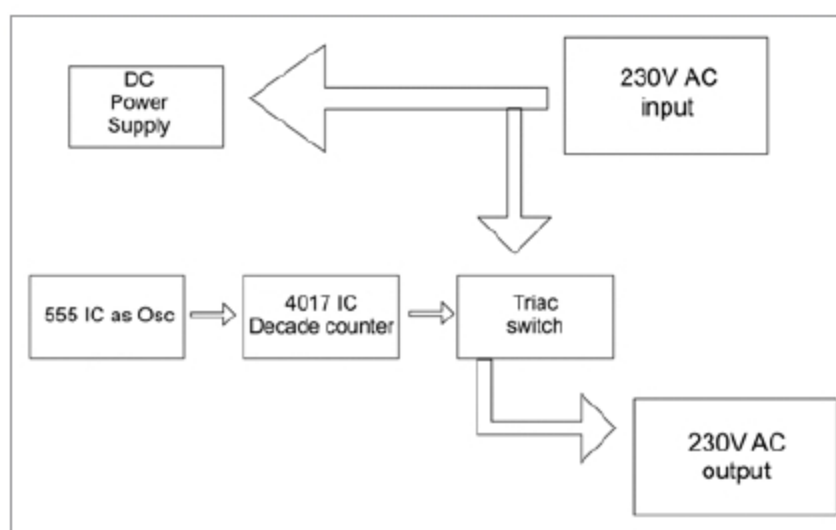


Fig. 2: Block diagram of the power-on delay timer

decade counter 4017B (IC2), optotriac MOC3041 (IC3), triac BT139 (TRIAC1), a 100-kilo-ohm preset (VR1) and a few other components.

Timing accuracy is achieved by using decade counter 4017B. NE555 IC is used as a bistable oscillator. VR1 preset is used to vary the timings from 0.1-sec to 1-sec pulse in NE555 IC. When the main power is switched on, 4017B starts counting the pulses received from NE555. The output pulses from NE555 are fed to clock input pin 14 of 4017B. As a result, 4017B gets breezed and its ten outputs (Q0 through Q9) become high sequentially. Output pin 11 (Q9) of 4017B goes high at 10th pulse received from NE555.

The output is built around triac BT139, which can switch up to 16-ampere AC load. The maximum 16-ampere current means we can have 60 per cent as optimum current, that is, up to 9.6 amperes (or $16 \times 0.6 = 9.6$ amperes). The optimum load will be $220V \times 9.6A = 2112$ watts. The triac is trig-

gered by optotriac MOC3041 (IC3). Optotriac has the advantage of triggering 230V AC power by itself and providing circuit isolation.

DC power for this circuit is derived from step-down transformer X1, bridge rectifier BR1 and capacitors C3 and C4. Resistor R2, capacitor C5 and preset VR1 are timing components for NE555 IC. You can change the delay time (0.1 to 1-sec) of this circuit by varying VR1.

During power-on, clock enable pin 13 (CLK EN) and pin 11 (Q9) of 4017B are in low states. Each output of 4017B goes high one by one after each clock pulse. When Q9 goes high with tenth pulse, CLK EN also goes high. This stops the counting process in

PARTS LIST

Semiconductors:

IC1	- NE555 timer
IC2	- 4017B decade counter
IC3	- MOC3041 optotriac
Triac1	- BT139/BTA12 triac
BR1	- 1A, bridge rectifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

R1	- 100-ohm, 0.5W
R2-R5	- 10-kilo-ohm
R6	- 330-ohm, 0.5W
R7	- 1.5-kilo-ohm
R8	- 150-ohm, 0.5W
R9	- 1-mega-ohm
VR1	- 100-kilo-ohm preset

Capacitors:

C1	- 100 μ F, 25V electrolytic
C2, C3, C6	- 0.1 μ F ceramic disk
C4	- 1000 μ F, 25V electrolytic
C5	- 4.7 μ F, 25V electrolytic
C7	- 0.01 μ F, 400V ceramic disk

Miscellaneous:

CON1, CON2	- 3-pin terminal connector
MOV1	- 270V metal oxide varistor
X1	- 230V AC primary to 9V, 300mA secondary transformer
	- Heat-sink for triac

(All these are available from kitsnspares.com)